

# TITAN research-software demonstration

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## Research-software output: internally derived TCGA estimates

### COAD - TCGA-AA-A01F

| Models applied | Continuous | Binary | Slides pooled |
|----------------|------------|--------|---------------|
| 20             | 10         | 10     | 1             |

This research-software demonstration applies models developed and evaluated only in TCGA. Every displayed value is an **internally derived TCGA estimate**, not external performance evidence or a diagnostic output. Binary LDA scores are uncalibrated. Their displayed TCGA out-of-fold score ranks describe relative position only and must not be read as probabilities.

### Prominent tissue-source-site sensitivity warning

**12 model(s) in this output fell below their original effect threshold under TCGA tissue-source-site-grouped internal validation. These calls must not be highlighted without this warning.** Tissue-source-site grouping is not institutional or scanner-level external validation.

| Endpoint (type)                                      | TSS-grouped metric | Grouped minus random | Warning category       |
|------------------------------------------------------|--------------------|----------------------|------------------------|
| APC (binary)                                         | BA 0.543           | -0.250               | Near chance or worse   |
| PI3K (binary)                                        | BA 0.593           | -0.020               | Below effect threshold |
| Aneuploidy score (continuous)                        | Q2 0.173           | -0.086               | Below effect threshold |
| Deleted arm count (continuous)                       | Q2 0.147           | -0.066               | Below effect threshold |
| MANTIS score (continuous)                            | Q2 0.113           | -0.104               | Below effect threshold |
| MSIsensor score (continuous)                         | Q2 0.026           | -0.229               | Below effect threshold |
| Aneuploidy Score (continuous)                        | Q2 0.174           | -0.096               | Below effect threshold |
| Lymphocyte Infiltration Signature Score (continuous) | Q2 0.048           | -0.238               | Below effect threshold |
| Nonsilent Mutation Rate (continuous)                 | Q2 0.088           | -0.127               | Below effect threshold |
| SNV Neoantigens (continuous)                         | Q2 -0.013          | -0.252               | Near chance or worse   |
| Silent Mutation Rate (continuous)                    | Q2 -0.003          | -0.236               | Near chance or worse   |
| TIL Regional Fraction (continuous)                   | Q2 0.176           | -0.278               | Below effect threshold |

### Case context (not used as model input)

**Shared selection features.** Both cases were men with sigmoid-colon, moderately differentiated (G2), pT3 adenocarcinoma and tumour-free resection margins.

**Pathology summary.** The sigmoid resection contained an ulcerated grade-2 adenocarcinoma that had crossed the muscular wall into pericolic fat (pT3). Both resection ends were free of carcinoma; lymphatic invasion was recorded, and 2 of 30 regional nodes contained metastasis (pN1).

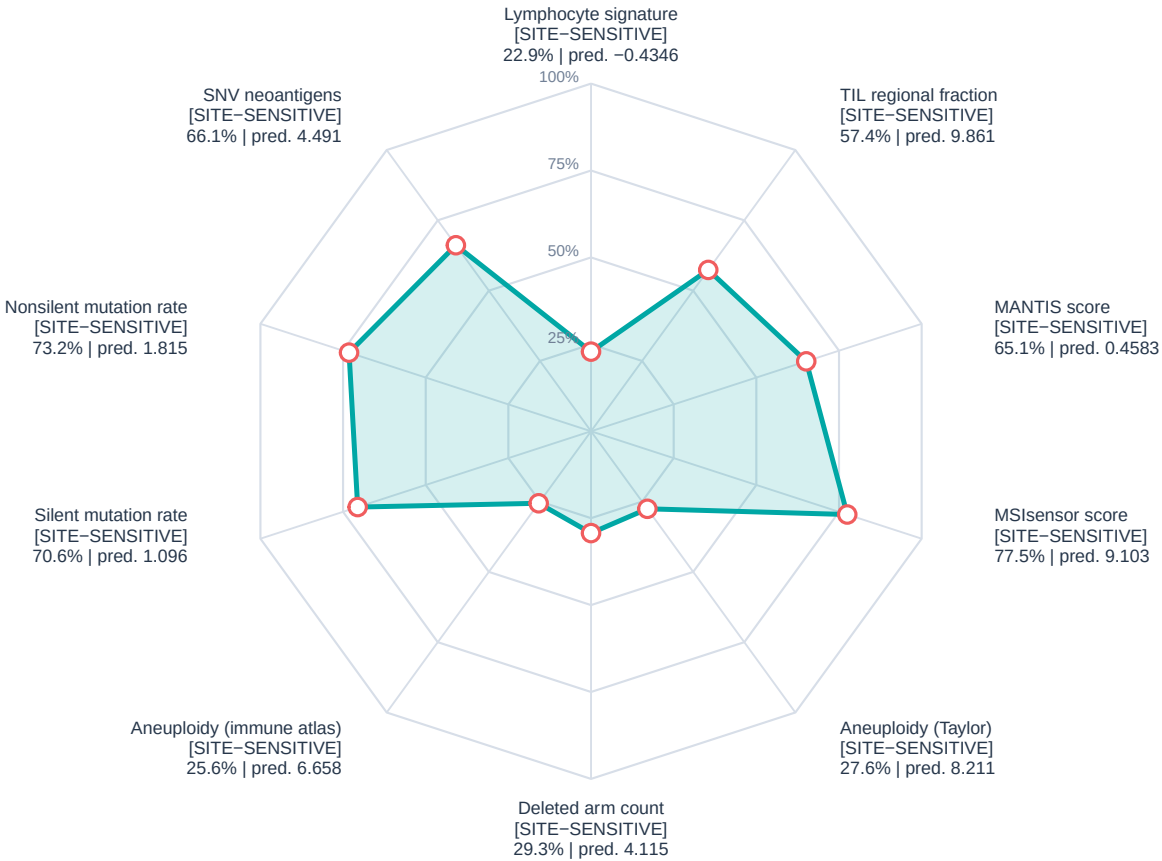
**Treatment and follow-up.** NCI GDC records list fluorouracil, leucovorin calcium and oxaliplatin from day 62 to day 215, each with the recorded outcome ‘Complete Response’. Follow-up disease status was ‘with tumor’ on day 31 and ‘unknown’ on day 974; the recorded vital status was alive.

**Clinical-context source.** Pathology was paraphrased from TCGA-Slide-Reports.csv distributed with the TITAN study (Ding et al., Nature Medicine 2025; doi:10.1038/s41591-025-03982-3). Treatment and follow-up fields were retrieved from the public NCI GDC Cases API (<https://api.gdc.cancer.gov/cases>) on 16 August 2026.

The pathology and treatment fields are descriptive public metadata; they were not predictors, training targets, or validation outcomes for TITANPred. No treatment-effect inference should be drawn from these examples.

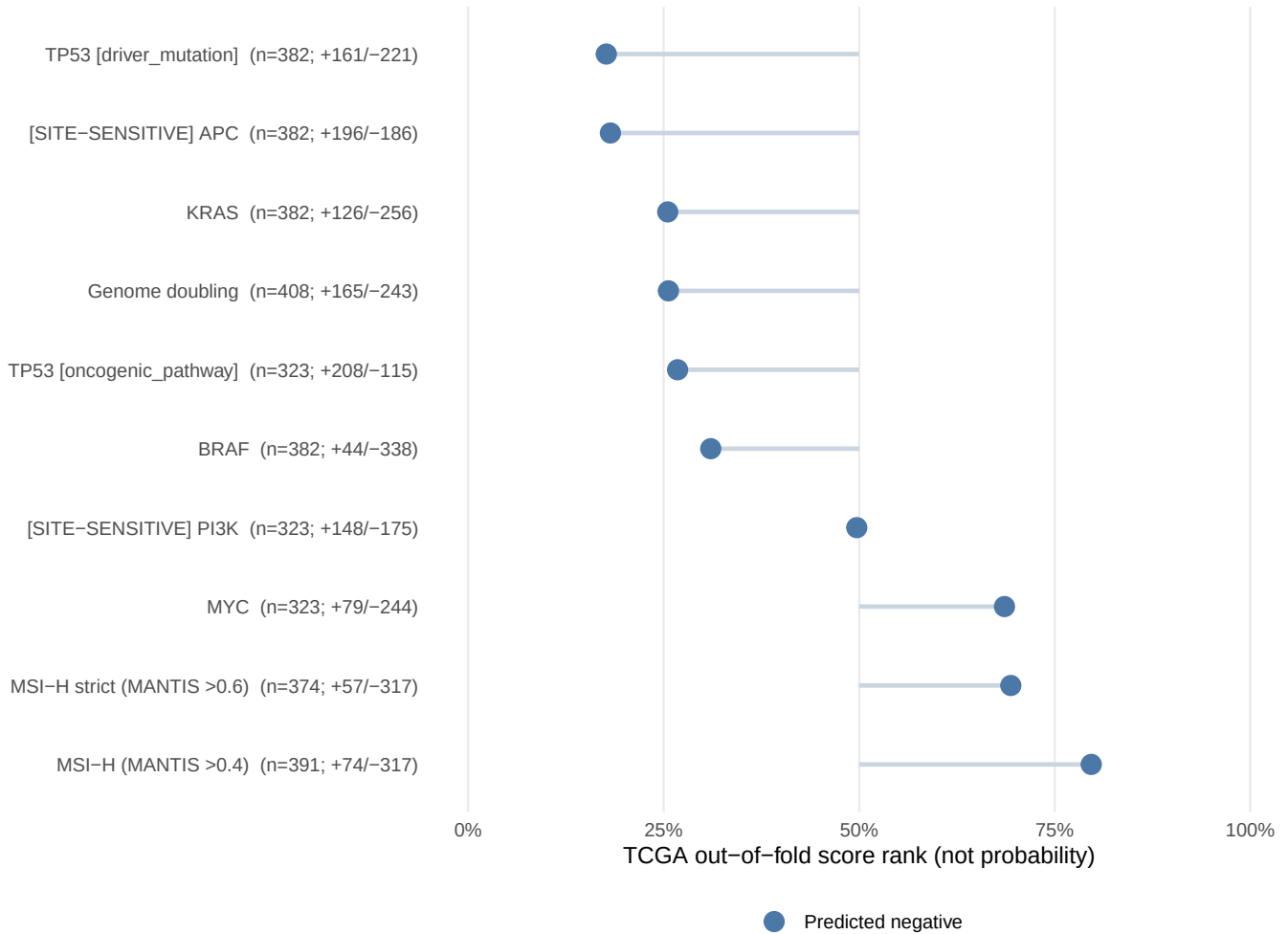
### Continuous internally derived TCGA estimates

[SITE-SENSITIVE] marks models below the original effect threshold under TCGA TSS grouping



## Binary research-model calls

TCGA out-of-fold LDA-score rank; [SITE-SENSITIVE] models require prominent c:



## Continuous internally derived estimates

| Family                     | Endpoint                                | Prediction | Reference percentile | Category        |
|----------------------------|-----------------------------------------|------------|----------------------|-----------------|
| microsatellite_instability | MSIsensor score                         | 9.103      | 77.5%                | Moderate effect |
| thorsson                   | Lymphocyte Infiltration Signature Score | -0.4346    | 22.9%                | Moderate effect |
| thorsson                   | Aneuploidy Score                        | 6.658      | 25.6%                | Moderate effect |
| thorsson                   | Nonsilent Mutation Rate                 | 1.815      | 73.2%                | Moderate effect |
| aneuploidy                 | Aneuploidy score                        | 8.211      | 27.6%                | Moderate effect |
| aneuploidy                 | Deleted arm count                       | 4.115      | 29.3%                | Moderate effect |
| thorsson                   | Silent Mutation Rate                    | 1.096      | 70.6%                | Moderate effect |
| thorsson                   | SNV Neoantigens                         | 4.491      | 66.1%                | Moderate effect |
| microsatellite_instability | MANTIS score                            | 0.4583     | 65.1%                | Moderate effect |
| thorsson                   | TIL Regional Fraction                   | 9.861      | 57.4%                | Higher effect   |

## Binary research-model calls

| Endpoint                   | Predicted class | LDA score | TCGA OOF score rank (not probability) | Training n | Positiv |
|----------------------------|-----------------|-----------|---------------------------------------|------------|---------|
| Genome doubling            | 0               | -2.153    | 25.6%                                 | 408        | 16      |
| APC                        | 0               | -2.982    | 18.2%                                 | 382        | 19      |
| BRAF                       | 0               | -6.932    | 31.0%                                 | 382        | 4       |
| KRAS                       | 0               | -3.159    | 25.5%                                 | 382        | 12      |
| TP53 [driver_mutation]     | 0               | -3.171    | 17.7%                                 | 382        | 16      |
| MSI-H (MANTIS >0.4)        | 0               | -1.165    | 79.7%                                 | 391        | 7       |
| MSI-H strict (MANTIS >0.6) | 0               | -4.352    | 69.4%                                 | 374        | 5       |
| MYC                        | 0               | -1.023    | 68.6%                                 | 323        | 7       |
| PI3K                       | 0               | -0.2634   | 49.7%                                 | 323        | 14      |
| TP53 [oncogenic_pathway]   | 0               | -0.05124  | 26.8%                                 | 323        | 20      |

## Quality and provenance

- Patient-level aggregation: arithmetic mean across all supplied slide vectors.
- Maximum fraction outside endpoint-specific TCGA training ranges: 0.3%.
- TITAN input schema: 768 named features (`titan_000–titan_767`).
- Decomposition: seeded rSVD in fastPLS; binary classification: PLS-LDA.
- Tissue-source-site metadata: every row includes the grouped internal metric, random-minus-grouped change, analysed-site count, threshold-retention status, fold-size range and inner-site-separation audit.
- Scope: TCGA tissue-source-site-grouped internal validation is not institutional or scanner-level external validation.
- Intended use: research-software demonstration only; TCGA benchmark without independent external validation.

## Endpoint interpretation and target sources

For continuous endpoints, the percentile locates the prediction within repeated TCGA out-of-fold predictions and is not a clinical reference interval. For binary endpoints, the displayed value is the rank of the uncalibrated LDA score among repeated TCGA out-of-fold scores. It is not a probability, confidence, or clinical-risk estimate. Training denominators and repeated-CV performance are shown beside each binary call; rows marked for limited class size require particular caution.

- **Genome doubling.** Binary whole-genome-doubling call derived from the tumour copy-number profile. *Analysed scale:* PLS-LDA class and uncalibrated discriminant score *Model-target source:* Taylor et al., Cancer Cell 2018, Table S2; doi:10.1016/j.ccell.2018.03.007.
- **APC.** Binary prediction of a qualifying protein-altering PASS mutation in APC; the displayed LDA score is not a probability. *Analysed scale:* PLS-LDA class and uncalibrated discriminant score *Model-target source:* MC3 consensus calls: Ellrott et al., Cell Systems 2018, doi:10.1016/j.cels.2018.03.002; cancer-gene eligibility: Bailey et al., Cell 2018, doi:10.1016/j.cell.2018.02.060.
- **BRAF.** Binary prediction of a qualifying protein-altering PASS mutation in BRAF; the displayed LDA score is not a probability. *Analysed scale:* PLS-LDA class and uncalibrated discriminant score *Model-target source:* MC3 consensus calls: Ellrott et al., Cell Systems 2018, doi:10.1016/j.cels.2018.03.002; cancer-gene eligibility: Bailey et al., Cell 2018, doi:10.1016/j.cell.2018.02.060.
- **KRAS.** Binary prediction of a qualifying protein-altering PASS mutation in KRAS; the displayed LDA score is not a probability. *Analysed scale:* PLS-LDA class and uncalibrated discriminant score *Model-target source:* MC3 consensus calls: Ellrott et al., Cell Systems 2018, doi:10.1016/j.cels.2018.03.002; cancer-gene eligibility: Bailey et al., Cell 2018, doi:10.1016/j.cell.2018.02.060.
- **TP53 [driver\_mutation].** Binary prediction of a qualifying protein-altering PASS mutation in TP53; the displayed LDA score is not a probability. *Analysed scale:* PLS-LDA class and uncalibrated discriminant score *Model-target source:* MC3 consensus calls: Ellrott et al., Cell Systems 2018, doi:10.1016/j.cels.2018.03.002; cancer-gene eligibility: Bailey et al., Cell 2018, doi:10.1016/j.cell.2018.02.060.
- **MSI-H (MANTIS >0.4).** Binary call indicating whether the source MANTIS score exceeds 0.4; the displayed LDA score is not a probability. *Analysed scale:* PLS-LDA class and uncalibrated discriminant score *Model-target source:* Bonneville et al., JCO Precision Oncology 2017; doi:10.1200/PO.17.00073.

- **MSI-H strict (MANTIS >0.6).** Sensitivity-analysis call: MANTIS >0.6 is positive, <0.4 is negative, and intermediate source values were excluded from training. *Analysed scale:* PLS-LDA class and uncalibrated discriminant score *Model-target source:* Threshold documentation distributed with the cBioPortal TCGA PanCancer Atlas clinical sample files.
- **MYC.** Binary prediction that the MYC oncogenic pathway carries at least one qualifying genomic alteration; the displayed LDA score is not a probability. *Analysed scale:* PLS-LDA class and uncalibrated discriminant score *Model-target source:* Sanchez-Vega et al., Cell 2018, Table S4; doi:10.1016/j.cell.2018.03.035.
- **PI3K.** Binary prediction that the PI3K oncogenic pathway carries at least one qualifying genomic alteration; the displayed LDA score is not a probability. *Analysed scale:* PLS-LDA class and uncalibrated discriminant score *Model-target source:* Sanchez-Vega et al., Cell 2018, Table S4; doi:10.1016/j.cell.2018.03.035.
- **TP53 [oncogenic\_pathway].** Binary prediction that the TP53 oncogenic pathway carries at least one qualifying genomic alteration; the displayed LDA score is not a probability. *Analysed scale:* PLS-LDA class and uncalibrated discriminant score *Model-target source:* Sanchez-Vega et al., Cell 2018, Table S4; doi:10.1016/j.cell.2018.03.035.
- **Aneuploidy score.** Number of chromosome arms with arm-level copy-number gain or loss; larger values indicate greater chromosomal imbalance. *Analysed scale:* source endpoint units *Model-target source:* Taylor et al., Cancer Cell 2018, Table S2; doi:10.1016/j.ccell.2018.03.007.
- **Deleted arm count.** Number of chromosome arms classified as deleted; larger values indicate a greater arm-level deletion burden. *Analysed scale:* source endpoint units *Model-target source:* Taylor et al., Cancer Cell 2018, Table S2; doi:10.1016/j.ccell.2018.03.007.
- **MANTIS score.** Genome-wide microsatellite-instability score from the MANTIS algorithm; larger values indicate stronger MSI evidence. *Analysed scale:* source endpoint units *Model-target source:* Kautto et al., Oncotarget 2017; doi:10.18632/oncotarget.20432. TCGA values were obtained from cBioPortal PanCancer Atlas records.
- **MSIsensor score.** Microsatellite-instability score from MSIsensor; larger values indicate a larger fraction of unstable microsatellite loci. *Analysed scale:* source endpoint units *Model-target source:* Niu et al., Bioinformatics 2014; doi:10.1093/bioinformatics/btt755. TCGA values were obtained from cBioPortal PanCancer Atlas records.
- **Aneuploidy Score.** Aneuploidy burden distributed with the TCGA immune landscape resource; larger values indicate more arm-level copy-number alterations. *Analysed scale:* source endpoint units *Model-target source:* Thorsson et al., Immunity 2018, PanImmune\_MS; doi:10.1016/j.immuni.2018.03.023.
- **Lymphocyte Infiltration Signature Score.** Gene-expression signature summarising lymphocyte-associated transcriptional infiltration; it is not a direct cell count. *Analysed scale:* source endpoint units *Model-target source:* Thorsson et al., Immunity 2018, PanImmune\_MS; doi:10.1016/j.immuni.2018.03.023.
- **Nonsilent Mutation Rate.** Rate of coding mutations that alter the encoded protein. TITANPred reports the model output on the analysis  $\log(1+x)$  scale. *Analysed scale:*  $\log_{10}$ -transformed source endpoint units *Model-target source:* Thorsson et al., Immunity 2018, PanImmune\_MS; doi:10.1016/j.immuni.2018.03.023.
- **SNV Neoantigens.** Predicted neoantigen burden arising from single-nucleotide variants. TITANPred reports the model output on the analysis  $\log(1+x)$  scale. *Analysed scale:*  $\log_{10}$ -transformed source endpoint units *Model-target source:* Thorsson et al., Immunity 2018, PanImmune\_MS; doi:10.1016/j.immuni.2018.03.023.
- **Silent Mutation Rate.** Rate of coding mutations that do not change the encoded amino acid. TITANPred reports the model output on the analysis  $\log(1+x)$  scale. *Analysed scale:*  $\log_{10}$ -transformed source endpoint units *Model-target source:* Thorsson et al., Immunity 2018, PanImmune\_MS; doi:10.1016/j.immuni.2018.03.023.
- **TIL Regional Fraction.** Estimated fraction of tumour image regions containing tumour-infiltrating lymphocytes in the TCGA immune landscape resource. *Analysed scale:* source endpoint units *Model-target source:* Thorsson et al., Immunity 2018, PanImmune\_MS; doi:10.1016/j.immuni.2018.03.023.